



CJC
Complete Java Classes

OVERLOADING (POLYMORPHISM) TASK: -

List of tasks related to Overloading (Polymorphism)–

Question 1. What is polymorphism? How many ways to achieve polymorphism?

Question 2. Define Overloading. What are advantages and example of overloading?

Question 3. Explain the syntax rules of overloading.

Question 4. Can we overload static, private and final method? If yes then how?

Question 5. Can we overload java main method? If yes then how?

Question 6. The program for writing different ways of overloading methods.



```
public class Calculator {
```

```
    //number of parameters:
```

```
    public void calculate(int num1, int num2) {
```

```
        System.out.println("With two no of parameters:  
"+num1+num2);
```

```
    }
```

```
    public void calculate (int num1, int num2, int num3){
```

```
        System.out.println("With three no of parameters:  
"+num1+num2+num3);
```

```
    }
```

```
    //Type of parameters:
```

```
    public void calculate (float num1, float num2){
```

```
        System.out.println("With two float and float  
parameters: "+num1+num2);
```

```
    }
```

```
    public void calculate(double num1, double num2){
```

```
        System.out.println("With two double and double  
parameters: "+num1+num2);
```

```
    }
```



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//Order of parameters:

```
public void calculate(double num1, int num2){  
    System.out.println("With two different parameters:  
"+num1+num2);  
}  
  
public void calculate(int num1, double num2) {  
    System.out.println("With two different sequence parameters:  
"+num1+num2);  
}  
  
public static void main (String[] args){  
    Calculator c = new Calculator();  
    c.calculate(10, 20);  
    c.calculate(10,20,30);  
    c.calculate(10.1f,20.1f);  
    c.calculate(20.2d,30.2d);  
    c.calculate(10.1d,40);  
    c.calculate(10,20.1d);  
}  
}
```



Question 7. What is the output of below program.

```
public class A{  
    public void display(){  
        System.out.println("display_method");  
    }  
    public int display(int x){  
        return x;  
    }  
    public void display(String msg){  
        System.out.println("display_String_method msg: "+msg);  
    }  
    public double display(double a,double b){  
        return a+b;  
    }  
    public static void main(String[] args) {  
        A a=new A();  
        int x=a.display(20);  
    }  
}
```



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```
System.out.println("display_int_method output:"+x);  
    double y = a.display(10.1d, 30.2 d);  
System.out.println("display_double_method output: "+y);  
    a.display("Hello");  
    a.display();  
}  
}
```

Question 8. Can we overload main method () in Java? If yes then check the output of below program.

```
public class Test {  
    public static void main(String[] args){  
        System.out.println("Normal main method");  
        main("cjc");  
    }  
    public static void main(String arg1){  
        System.out.println("without array main method: " + arg1);  
        main("cjc", "karvenagar");  
    }  
    public static void main(String arg1, String arg2) {
```



```
System.out.println("two paramerter main method: "+ arg1  
+", "+arg2);  
}  
}
```

Question 9. Check the program correct or not. If yes What will be the output of the program?

```
public class Check {
```

```
    public void disp(int a, double b){  
        System.out.println("Method A");  
    }
```

```
    public void disp(int a, double b, double c){  
        System.out.println("Method B");  
    }
```

```
    public void disp(int a, int b){  
        System.out.println("Method C");  
    }
```

```
    public static void main(String args[]){  
        Check c= new Check();
```



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```
c.disp(100, 20.67f);  
    }  
}
```

Question 10. Check below program is correct or not. If not explain why?

```
public class CheckOverloading {
```

```
    public void mt () {
```

```
    }
```

```
    public void mt (int x) {
```

```
    }
```

```
    public int mt (int y) {
```

```
        return 10;
```

```
    }
```

```
    public void mt (double x) {
```

```
    }
```



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```
public void mt (int x, int y) {  
  
}  
public String mt () {  
  
    return "cjc";  
}  
}
```

Question 11. Write the program for Constructor Overloading.

THANK YOU